Bachelor's Degree
in Physics

Two-dimensional self-avoiding walks

Direct rejection and the Pivot algorithm

Candidate

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Supervisor

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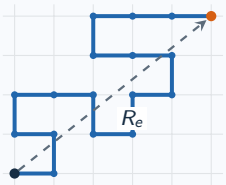
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Self-avoidance and polymer scaling

A **self-avoiding walk** never revisits a lattice site. It models excluded volume in a good solvent.

R_e is one walk's end-to-end distance; R is a trial chain size, whose Flory optimum satisfies $R_* \sim \sqrt{\langle R_e^2 \rangle}$.
d is the spatial dimension and $v > 0$ measures excluded-volume repulsion.

A self-avoiding chain



N steps of length a

$R_e^2 = |\mathbf{r}_N - \mathbf{r}_0|^2$

Entropic elasticity

$$\frac{F_{\mathrm{el}}}{k_B T} \sim \frac{R^2}{Na^2}$$

Stretching reduces entropy.

Excluded volume

$$\frac{F_{\mathrm{int}}}{k_B T} \sim \frac{vN^2}{R^d}$$

Crowding reduces the number of allowed conformations.

Flory approximation

$$\frac{F_{\mathrm{F}}(R)}{k_{\mathrm{B}} T} \sim \frac{R^2}{Na^2} + \frac{vN^2}{R^d}$$

Minimise in $R \quad \Longrightarrow \quad R_* \sim aN^{\nu_F}, \qquad \nu_F = \dfrac{3}{d+2}$

Gaussian elasticity, uniform density, leading pair repulsion.
Asymptotic scaling for $N \gg 1$ with $v \sim a^d > 0$.

In two dimensions: $\nu_F = \frac{3}{4} \quad \Longrightarrow \quad \langle R_e^2 \rangle \sim R_*^2 \sim a^2 N^{3/2}$ $\qquad a = 1$

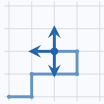
Simple sampling by direct rejection

Grow independent lattice walks; reject the whole trial at the first repeated site.

1 Start

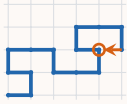
Start at the origin.
Mark it as visited.

2 Propose a step



No reversal: $1/3$ per direction.
A reversal fails the site check.

3 Check the site



New site: continue.
Visited site: reject and restart.

4 Accept and measure



After N valid steps, accept the walk.

$R_e^2 = |\mathbf{r}_N - \mathbf{r}_0|^2$

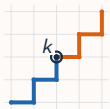
Observables and scaling $\langle R_e^2 \rangle \propto N^{2\nu}$ P_{success} ν (fit)

The Pivot algorithm: global shape updates

Start from a straight SAW. A rigid lattice symmetry reshapes the tail of the current walk.
Each move preserves all N lattice bonds. Successive configurations remain correlated.

One Pivot attempt

1 Choose a pivot



Choose k at random.
Prefix and tail.

2 Transform tail



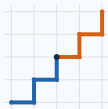
Fix the prefix.
Rotate or reflect.

3 Check overlap



Check the new tail
against the prefix.

4 Accept or reject



Valid: accept.
Collision: reject.

Repeat from the current walk

Seven equally likely moves: rotations by 90°, 180°, 270° and reflections in axes or diagonals.



Thermalisation

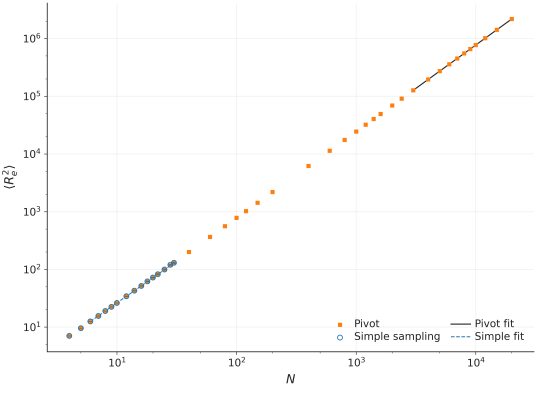
Repeat steps 1–4.
Discard initial attempts
to reduce the starting bias.

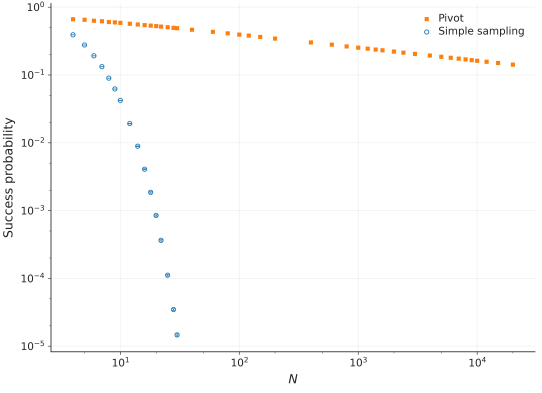


Production

Continue steps 1–4.
Record R_e^2 every Δt attempts.
Include rejected moves.

Chain extension and sampling success





Simple sampling

$\nu = 0.724 \pm 0.002$

$\chi^2/\mathrm{ndof} = 1.53 \quad (\mathrm{ndof} = 12)$

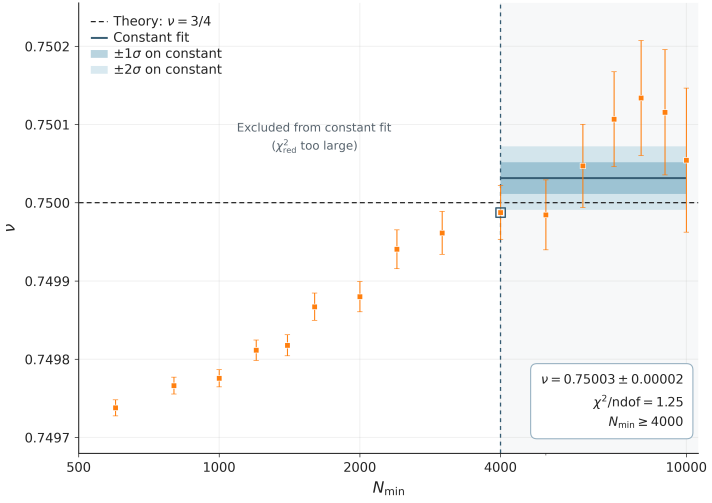
Pivot

$\nu = 0.74996 \pm 0.00003$

$\chi^2/\mathrm{ndof} = 1.48 \quad (\mathrm{ndof} = 9)$

Run time approximately 5 days and 16 hours

Scaling stability and future directions



Pivot: power-law fits using $N \geq N_{\min}$

Future directions

- **Three dimensions**

  SAWs in 3D, closer to real polymer geometry.

- **Interacting SAWs**

  Add attractive contacts between nonconsecutive neighbours (ISAW).

  Vary solvent quality: swollen or compact chains.

References

[1] P.-G. de Gennes, *Scaling Concepts in Polymer Physics*, Cornell University Press, Ithaca (1979).

[2] K. Binder and D. W. Heermann, *Monte Carlo Simulation in Statistical Physics: An Introduction*, 4th ed., Springer, Berlin (2002).

[3] N. Madras and A. D. Sokal, "The Pivot Algorithm: A Highly Efficient Monte Carlo Method for the Self-Avoiding Walk", *Journal of Statistical Physics* **50**, 109–186 (1988).

[4] S. Caracciolo, A. J. Guttmann, I. Jensen, A. Pelissetto, A. N. Rogers and A. D. Sokal, "Correction-to-Scaling Exponents for Two-Dimensional Self-Avoiding Walks", *Journal of Statistical Physics* **120**, 1037–1100 (2005).

Backup

Flory free energy: entropy and excluded volume

N bonds of length a, spatial extent $R = |\mathbf{R}|$, and $1 < d < 4$.
Long-chain, good-solvent regime: $N + 1 \simeq N \gg 1$ monomers and $v > 0$.

Gaussian elastic cost

$$P_0(\mathbf{R}) \simeq \left(\frac{d}{2\pi N a^2}\right)^{d/2} \exp\left[-\frac{d R^2}{2 N a^2}\right]$$

$$\frac{F_{\mathrm{el}}}{k_{\mathrm{B}} T} = -\ln P_0 + \mathrm{const} = C_{\mathrm{el}} \frac{R^2}{N a^2}$$

Ideal-chain reference: stretching reduces entropy.

Pair excluded volume

$$V = \kappa_d R^d, \qquad \rho \simeq \frac{N}{V}$$

$$\frac{F_{\mathrm{rep}}}{k_{\mathrm{B}} T} \simeq \frac{v}{2} \int \rho(\mathbf{r})^2 \, \mathrm{d}^d r = \frac{v N^2}{2 V}$$

$$\frac{F_{\mathrm{rep}}}{k_{\mathrm{B}} T} = C_{\mathrm{rep}} \frac{v N^2}{R^d}$$

Approximations: Gaussian elasticity, uniform density and local pair repulsion.
Spatial correlations and higher density terms are omitted. v has units of length^d.

$$\Phi(R) \equiv \frac{F_{\mathrm{F}}(R)}{k_{\mathrm{B}} T} = C_{\mathrm{el}} \frac{R^2}{N a^2} + C_{\mathrm{rep}} \frac{v N^2}{R^d}$$

$C_{\mathrm{el}} = d/2$, $\quad C_{\mathrm{rep}} = 1/(2\kappa_d) > 0$: prefactors do not depend on N or R.

de Gennes (1979); Bazant, Polymer Models, Sec. 3.2.

From the free-energy minimum to the power law

Minimum of the Flory free energy

$$\Phi'(R_*) = \frac{2C_{\mathrm{el}}R_*}{Na^2} - \frac{dC_{\mathrm{rep}}vN^2}{R_*^{d+1}} = 0 \quad \Longrightarrow \quad R_*^{d+2} = \frac{dC_{\mathrm{rep}}}{2C_{\mathrm{el}}}\, va^2N^3$$

$$v \sim a^d \quad \Longrightarrow \quad R_* \sim aN^{\nu_{\mathrm{F}}}, \qquad \nu_{\mathrm{F}} = \frac{3}{d+2}$$

$$\Phi''(R) = \frac{2C_{\mathrm{el}}}{Na^2} + \frac{d(d+1)C_{\mathrm{rep}}vN^2}{R^{d+2}} > 0 \quad (R > 0): \text{ unique minimum}$$

Second moment: assume one characteristic length

$$P_N(\mathbf{R}) \simeq \ell_N^{-d}\, p(\mathbf{R}/\ell_N), \quad \ell_N = aBN^{\nu}, \quad \int p(\mathbf{x})\,\mathrm{d}^d x = 1$$

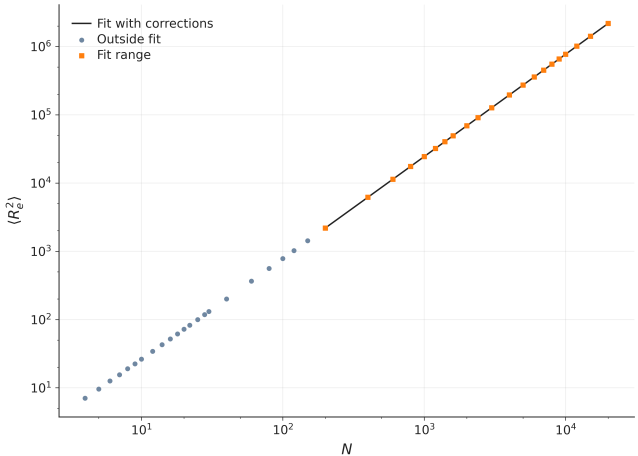
$$\langle R_e^2 \rangle = \int R^2 P_N(\mathbf{R})\,\mathrm{d}^d R \simeq \ell_N^2 \underbrace{\int |\mathbf{x}|^2 p(\mathbf{x})\,\mathrm{d}^d x}_{C_2 < \infty} = AN^{2\nu}$$

$A = a^2B^2C_2$ $\qquad$ $d = 2: \quad \nu_{\mathrm{F}} = 3/4 \quad \Longrightarrow \quad \langle R_e^2 \rangle \sim AN^{3/2}$

Conditional derivation under Flory/scaling assumptions, not a rigorous proof for lattice SAWs.
Caracciolo et al. (2005), Sec. 2; Bauerschmidt et al. (2012), Sec. 1.6: planar prediction $\nu = 3/4$.

Finite-size corrections: a broader fit range

$$\langle R_e^2 \rangle = A N^{2\nu} \left(1 + \frac{b}{N} + \frac{c}{N^{3/2}}\right)$$



Pivot fit results

$200 \leq N \leq 20000$

$\nu = 0.75005 \pm 0.00003$

$\chi^2/\mathrm{ndof} = 1.02 \quad (\mathrm{ndof} = 17)$

$A = 0.7704 \pm 0.0004$

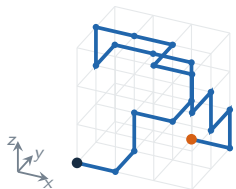
$b = 1.8 \pm 0.2$

$c = -6 \pm 2$

Four free parameters; fixed correction powers. Errors: one standard deviation.

Future models: three dimensions and attractive contacts

Three-dimensional SAWs

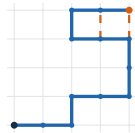


Cubic lattice: steps along $\pm x, \pm y, \pm z$.

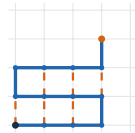
Use cubic-lattice symmetries for Pivot moves.

Numerical reference: $\nu_{3\mathrm{D}} \simeq 0.588$.
Flory estimates 3/5, illustrating its limits.

Interacting SAWs (ISAW)



Swollen: $m = 2$



Compact: $m = 6$

Same $N = 12$; dashed lines mark nonbonded contacts.

$$E = -\varepsilon m, \quad \varepsilon > 0, \qquad \pi(\omega) \propto e^{\beta\varepsilon m(\omega)}$$

m: nonbonded nearest-neighbour contact pairs.
Larger $\beta\varepsilon$ favours compact conformations.

$$P_{\mathrm{acc}} = \min(1, e^{-\beta\Delta E}), \quad \beta = (k_{\mathrm{B}}T)^{-1}$$

Symmetric valid proposals; reject all site overlaps.

Schematic configurations, not simulation results. Clisby, PRL 104, 055702 (2010); Hsu and Grassberger, J. Phys. A 35, L759 (2002).